



Arsi University

College of Health science

Department of pediatrics and child health

PERINATAL ASPHYXIA

By Tejitu U. (Pediatrician)

INTRODUCTION

- **PNA** results from **compromised placental** or **pulmonary gas exchange**.
- This disorder can lead to **hypoxia and hypercarbia** in the blood.
- **Severe hypoxia results in anaerobic glycolysis** and **lactic acid production** first in the peripheral tissues (muscle and heart) and then in the brain.

Terminology

- **Hypoxemia**:- decreased arterial concentration of oxygen.
- **Hypoxia**:- decreased oxygenation to cells or organs.
- **Ischemia**:- refers to blood flow to cells or organs that is insufficient to maintain their normal function.
- **Asphyxia** :- refers to progressive hypoxia, hypercarbia and acidosis.

Conti...

- **Neonatal encephalopathy**- a clinical term describing an abnormal neurobehavioral state (altered level of consciousness)
- **Hypoxic-ischemic encephalopathy (HIE)**- describes clinical evidence of encephalopathy a hypoxic-ischemic (HI) mechanism.

Neonatal encephalopathy

- A heterogeneous, clinically defined syndrome characterized by **disturbed neurologic function in the earliest days of life.**
- Manifested by a **reduced level of consciousness or seizures .**
- often accompanied by **difficulty with initiating and maintaining respiration.**
- Depression of tone and reflexes
- Causes **23%** of all neonatal deaths worldwide

ACOG definition/criteria for PNA

- Apgar score of **<5** at **5 minutes** and **10 minutes**.
- Fetal umbilical artery **pH <7.0**, **or** base deficit ≥ 12 mmol/L, **or** both
- Acute brain injury seen on **brain MRI** or magnetic resonance spectroscopy consistent with hypoxia–ischemia
- Presence of multisystem organ failure consistent with hypoxic–ischemic encephalopathy

Risk factors

Antepartum conditions

- Abnormal maternal oxygenation (eg, severe anemia, cardiopulmonary disease)
- Inadequate placental perfusion and/or gas exchange (eg, maternal hypertension or severe hypotension, placental insufficiency caused by vascular disease)
- Congenital infection or anomalies

Conti...

Intrapartum events

- Interruption of umbilical circulation
(eg, true knot, cord prolapse, cord avulsion)
- Inadequate placental perfusion and/or gas exchange
(eg, placental abruption, uterine rupture, severe maternal hypotension, abnormal uterine contractions)
- Traumatic delivery (eg, shoulder dystocia, difficult breech extraction)
- Abnormal maternal oxygenation (eg, pulmonary edema)

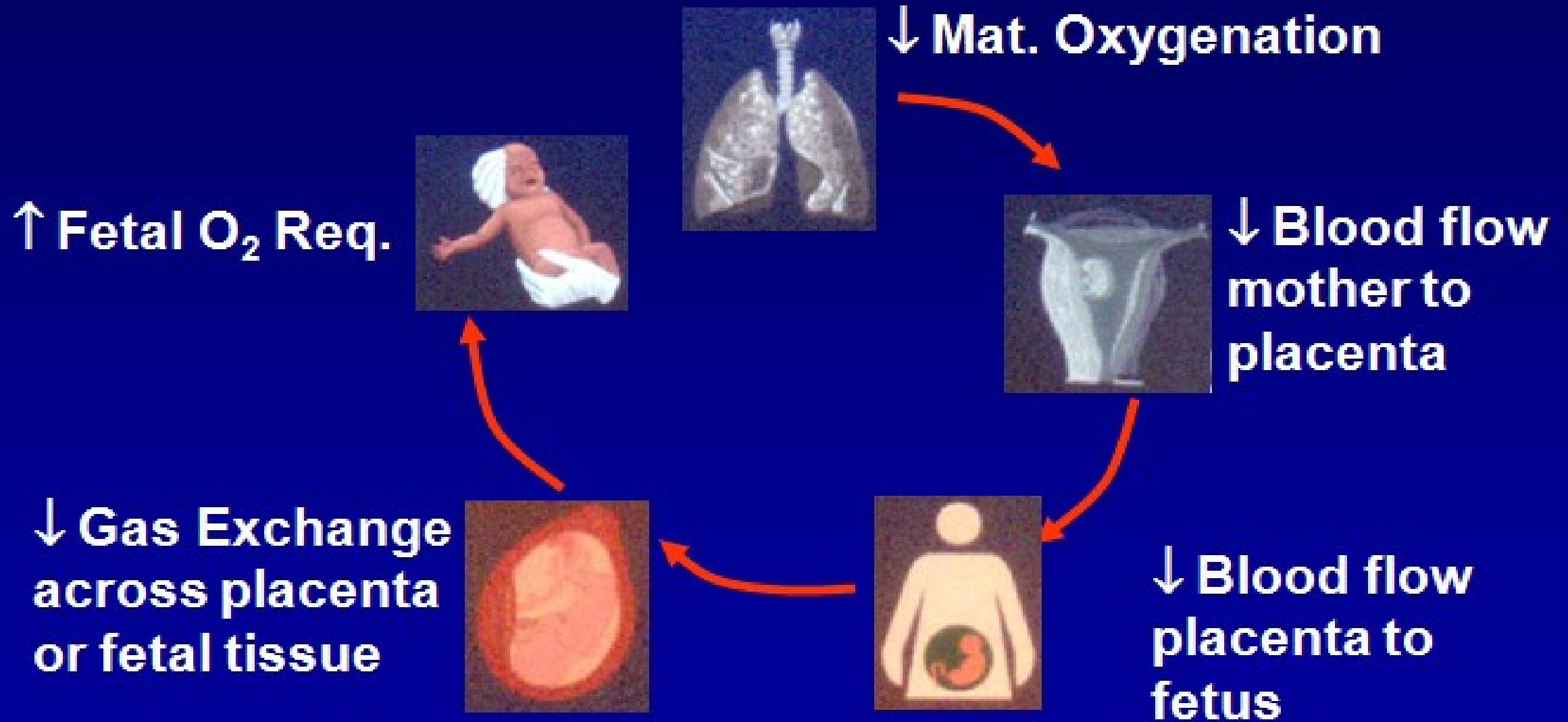
Conti...

Postnatal disorders (10%)

- ✓ Persistent pulmonary hypertension of the newborn (PPHN)
- ✓ Severe circulatory insufficiency (eg, acute blood loss, septic shock)
- ✓ Congenital heart disease

ETIOLOGY	
MATERNAL FACTORS	hypertension (acute or chronic), hypotension, infection (including chorioamnionitis), hypoxiafrom pulmonary or cardiac disorders, diabetes, maternal vascular disease, and in utero exposure to cocaine
PLACENTAL FACTORS	abnormal placentation, abruption, infarction, fibrosis, or hydrops
UTERINE RUPTURE	
UMBILICAL CORD ACCIDENTS	prolapse, entanglement, true knot, compression
ABNORMALITIES OF UMBILICAL VESSELS	
FETAL FACTORS	anemia (e.g., from fetal-maternal hemorrhage), infection, cardiomyopathy, hydrops, severe cardiac/circulatory insufficiency
NEONATAL FACTORS	cyanotic congenital heart disease, persistent pulmonary hypertension of the newborn (PPHN), cardiomyopathy, other forms of neonatal cardiogenic and/or septic shock, respiratory failure due tomeconium aspiration syndrome, neonatal pneumonia, pneumothorax,

Pathophysiology



After an episode of hypoxia and ischemia



Decreased ATP



Increased lactate and inorganic phosphates



Accumulation of toxic aminoacid like **glutamate**



Overactivation of NMDA, AMPA, and kainite receptors



Increased cellular permeability to sodium and calcium ions



cytotoxic edema and neuronal death

Mechanism of hypocalcemia in PNA

- Hyperphosphatemia:- Increased endogenous phosphate load due to tissue catabolism
- Delayed introduction of feeds
- Increased calcitonin production
- Renal failure
- Metabolic acidosis
- Its treatment with alkali therapy

Acute asphyxia

↓
Diving seal reflex

↓
Shunting of blood to brain, adrenals & heart

↓
Away from lungs, kidney, skin

NONE BRAIN ORGAN DAMAGE

Asphyxia continues

↓
Shunting within the brain

↓
Anterior Circulation Suffers

↓
Posterior Circulation Maintained

CEREBRAL CORTICAL LESIONS

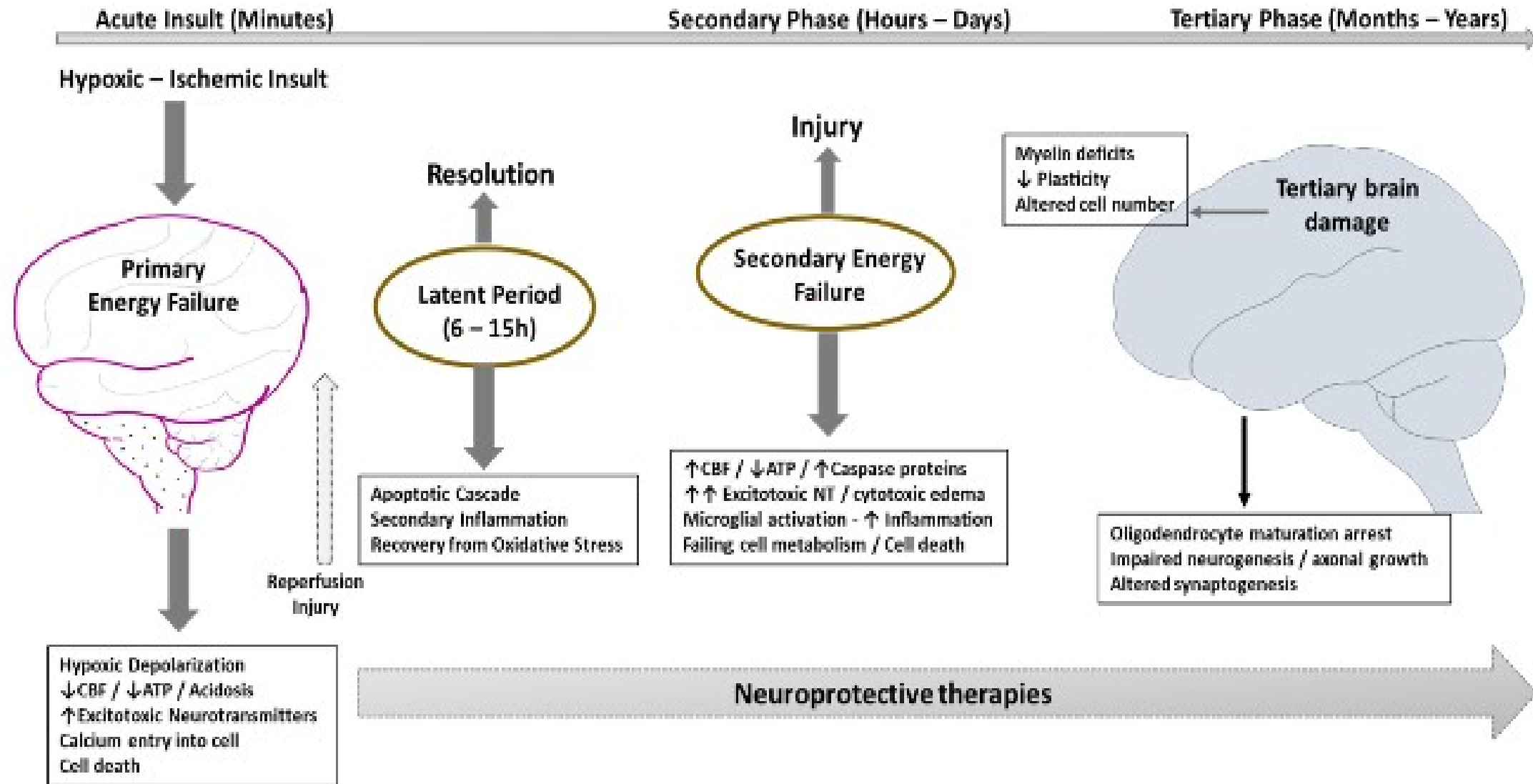


Figure 1. Schematic illustration of pathophysiology of HIE in relation to hypoxic ischemic (HI) insult resulting in primary (acute phase) and secondary energy failure (secondary phase) in the

Conti...

- The magnitude of the final neuronal damage depends
 - ➔ Initial insult
 - ➔ Damage due to energy failure
 - ➔ Reperfusion injury
 - ➔ Apoptosis

Pathology

➤ Target organs damage of perinatal asphyxia

- | | |
|-----------------------------|------|
| • Kidneys | 50% |
| • Brain | 28% |
| • Heart | 25% |
| • Lung | 23% |
| • Liver, Bowel, Bone marrow | < 5% |

Clinical Manifestations

- Asphyxia may be suspected and HIE
 - Prolonged (>1 hour) antenatal acidosis
 - Fetal HR <60 beats per minute
 - Apgar score ≤ 3 at ≥ 10 minutes
 - Need for PPV for >1 minute or first cry delayed >5minutes
 - Seizures within 12 to 24 hours of birth
 - **Burst suppression** or suppressed background pattern on EEG.

Conti...

- ✓ At birth, affected infants may have **neurologic impairment** and may **fail to breathe spontaneously**.
- ✓ The presence of **MSAF** is evident that fetal distress has occurred.
- ✓ Pallor, cyanosis, apnea, a slow heart rate, and unresponsiveness to stimulation are **initial signs of potential HIE**.

Conti...

- ✓ Infants may be **hypotonic**, may change from a hypotonic to a **hypertonic state**.
- ✓ **Cerebral edema** may develop during the next 24 hr and result in profound brainstem depression.
- ✓ **Seizure** activity may occur which may be severe and refractory to typical doses of anticonvulsants.

Conti...

➤ **Possible causes of seizure in PNA are:-**

- ✓ Hypoxic ischemic encephalopathy
- ✓ Cerebral edema
- ✓ Vascular events(hemorrhage, acute ischemic stroke, sinus venous thrombosis)
- ✓ Metabolic derangement (hypocalcemia, hypoglycemia)
- ✓ CNS infection

Multiorgan Systemic Effects of Asphyxia

SYSTEM	EFFECTS
Central nervous	Hypoxic-ischemic encephalopathy, infarction, intracranial hemorrhage, seizures, cerebral edema, hypotonia, hypertonia
Cardiovascular	Myocardial ischemia, poor contractility, cardiac stunning, tricuspid insufficiency, hypotension
Pulmonary	Pulmonary hypertension, pulmonary hemorrhage, respiratory distress syndrome
Renal	Acute tubular or cortical necrosis
Adrenal	Adrenal hemorrhage
Gastrointestinal	Perforation, ulceration with hemorrhage, necrosis
Metabolic	Inappropriate secretion of antidiuretic hormone, hyponatremia, hypoglycemia, hypocalcemia, myoglobinuria
Integumentary	Subcutaneous fat necrosis
Hematologic	Disseminated intravascular coagulation

Sarnat Stages of Hypoxic-Ischemic Encephalopathy

SIGNS	STAGE 1	STAGE 2	STAGE 3
Level of consciousness	Hyperalert	Lethargic	Stuporous, coma
Muscle tone	Normal	Hypotonic	Flaccid
Posture	Normal	Flexion	Decerebrate
Tendon reflexes/clonus	Hyperactive	Hyperactive	Absent
Myoclonus	Present	Present	Absent
Moro reflex	Strong	Weak	Absent
Pupils	Mydriasis	Miosis	Unequal, poor light reflex
Seizures	None	Common	Decerebration
Electroencephalographic findings	Normal	Low voltage changing to seizure activity	Burst suppression to isoelectric
Duration	<24 hr if progresses; otherwise, may remain normal	24 hr to 14 days	Days to weeks
Outcome	Good	Variable	Death, severe deficits

Laboratory Investigations

- ✓ Complete Blood Count
- ✓ Renal function tests
- ✓ Serum electrolytes
- ✓ Coagulation profiles
- ✓ Arterial Blood Gas analysis

Neuroimaging

➤ MRI:-

- ✓ The **most sensitive** imaging modality for detecting hypoxic brain injury in the neonate.
- ✓ Establish the presence and pattern of injury.
- ✓ Predict neurologic outcome/prognosis of the insult.

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- *MRI* findings
- After **severe acute total asphyxia**
 - ✓ Deep gray matter injury (putamen, thalamus)
 - ✓ Brainstem injury
- After **mild asphyxia**
 - ✓ Parasagittal injury of the cerebral cortex and
 - ✓ subcortical white matter in the arterial watershed distribution

Management

Supportive care

- ✓ Adequate resuscitation in delivery room
- ✓ Keep NPO
- ✓ Maintenance of adequate ventilation (avoidance of hypoxemia or hyperoxia)
- ✓ Maintenance of sufficient brain and organ perfusion (avoidance of systemic hypotension or hypertension)
- ✓ Control of seizures
- ✓ Avoid hyperthermia, Hypoglycemia, hyperglycemia

Feeding

- Keep NPO
- Put on calculated MF
- Start TF and escalate accordingly
- Add Ca in MF

Therapeutic hypothermia

- ✓ The only proven neuroprotective therapy for treatment of neonatal encephalopathy.
- ✓ Started within the first *6hrs* after delivery.
- ✓ Maintained *for 72 hrs* at 33 to 35°C with target temperature set at *33.5°C*.
- ✓ **2 Types**
 - ✓ Selective head cooling.
 - ✓ Whole body cooling.

Mechanism of action

1. **Decreased cerebral metabolism** and blood flow:

- *Decrease in energy requirement and*
- *cerebral edema*

2. **Decreased brain lactic acid, glutamate, and nitric oxide** concentrations:

- *Less excitotoxic and oxidative injury*

3. **Inhibits** protease activation, mitochondrial failure, free radical damage, lipid peroxidation:

- *Less apoptosis and necrosis*

4. **Reduced vascular permeability & disruptions of blood - brain barrier functions**

Criteria

- GA ≥ 36 wks and ≤ 6 hrs of age.
- A pH of ≤ 7.0 or a base deficit of ≥ 16 mmol/L
- One of the following:
 - ✓ A 10-minute Apgar score of < 5
 - ✓ Ongoing resuscitation initiated at birth and continued for at least 10 minutes
- Moderate to severe HIE on clinical examination

Contraindications

- ✓ Major congenital malformations
- ✓ Suspected/known chromosomal disorder
- ✓ Clinical and echocardiographic evidence of PPHN
- ✓ Active bleeding
- ✓ Major intracranial hemorrhage

Table 7.5: Monitoring (frequency)

Parameter	Day 1	Day 2	Day 3
Vitals monitoring including invasive blood pressure monitoring	Q 1 hour	Q 1 hour	Q 1 hour
Neurological monitoring	Q 12 hours	Q 12 hours	Q 12 hours
Urine output	Q 6 hours	Q 6 hours	Q 6 hours
ECG	Continuously	Continuously	Continuously
aEEG	Continuously	Continuously	Continuously
Skin integrity	Q 6 hours	Q 6 hours	Q 6 hours
Investigations			
Glucose	Q 6 hours	Q 6 hours	Q 6 hours
Blood gas	Q 6 hours or as indicated by condition of baby	Q 12 hours or as indicated	Q 12 hours or as indicated
Renal function test	Once	Once	Once
Electrolytes	Once	Once	Once
Complete hemogram	Only if required	Only if required	Only if required
Neurosonogram	If abnormal	If abnormal	If abnormal

Rewarming

- ✓ Increase the infant's core temperature by 0.5 C every hour until 36.5 C has been reached
- ✓ When the infant's core temperature is 36.5 C, remove the patient from the cooling blanket/device
- ✓ Re-activate the radiant warmer, monitor and document the infant's temperature with the skin probe
- ✓ Problems while rewarming:
 - **Seizures** and
 - **Hypotension**

Complications of induced hypothermia

- ✓ Thrombocytopenia
- ✓ Coagulopathy
- ✓ PPHN
- ✓ Hypotension
- ✓ Reduced heart rate
- ✓ Subcutaneous fat necrosis
- ✓ The potential for overcooling and the cold injury syndrome which is usually avoided with a servo-controlled cooling system

Management of seizure

- Ensure ABC
- Check **RBS** & manage **hypoglycemia**
- Second consider **hypocalcemia**
- Load calcium gluconate 200mg/kg iv with strict cardiac monitoring
- If seizure responds continue maintenance of calcium
- If no response with calcium gluconate loading consider **anti-epileptics**

➤ **Phenobarbital**

- ✓ The historical first line drug for neonatal seizures
- ✓ Loading dose (20 mg/kg).
- ✓ Additional doses of 5-10 mg/kg (up to 40 mg/kg total) may be needed

➤ **Phenytoin**

- Second line drug (20 mg/kg loading dose)

Duration of therapy

- There has been an increasing trend toward early discontinuation of antiseizure drugs
- Discontinue before or shortly after discharge from the hospital.

Prognosis

- The outcome correlates with the **timing and severity of the insult**.
- The prognosis varies depending on the severity of the insult and the treatment.
- Infants with initial cord or initial blood **pH <6.7** have a **90% risk for death or severe neurodevelopmental impairment at 18 mo of age**.

Poor Predictive Variables for Death/Disability After Hypoxic-Ischemic Encephalopathy

- Low (0–3) 10 min Apgar score
- Need for CPR in the delivery room
- Delayed onset (≥ 20 min) of spontaneous breathing
- Severe neurologic signs (coma, hypotonia, hypertonia)
- Seizures onset ≤ 12 hr or difficult to treat
- Severe, prolonged (~ 7 days) EEG findings including burst suppression pattern
- Prominent MRI basal ganglia/thalamic lesions
- Oliguria/anuria > 24 hr
- Abnormal neurologic exam ≥ 14 days

Follow up

- ❖ All survivors of HIE require comprehensive **high-risk medical and developmental follow-up.**
- ❖ The recommendations for follow-up and intervention are based on:-
 - ✓ *The clinical condition of the infant at the time of discharge*
 - ✓ *An assessment of feeding, vision, hearing*
 - ✓ *Whether seizures continue to be present or not*
 - ✓ *Head growth*
 - ✓ *General health and motor neurodevelopment*
 - ✓ *Assessments of cognitive skills and language development*

References

- Up-to-date 2018
- Nelson text book of pediatrics *21st edition*
- *Faranaof & marten's neonatal-perinatal medicine.*